Department of Computer Science and Engineering
Academic Year 2021– 2022(Even Semester)

Degree, Semester & Branch: B.E – II Semester CSE
Course Code & Title: CS3251 Programming in C
Name of the Faculty member (s): Ms.G.Sakthi Priya, AP/CSE

Innovative Practice Description

- **Unit / Topic: Unit IV / Structures**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO9
- **Activity Chosen: Code Finder**
- **Justification:**
  - Code Finder Activity: In this activity, the partially developed code segment will be given to the students. The students are asked to find the code to make it a full program
  - As the structure concept involve many sub topics like, Structure pointer, nested structure and array of structure, partially developed code of each topic is given to the students to make them understand each topic.
- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
  - Faculty prepared three programs from the topic Structure, Structure and pointer, Array of structures with partial code.
  - Faculty divided the class into 30 teams. Each team with size of two members.
  - The prepared materials were given to the student teams and asked to write the missed part of a program as shown in Fig 1
  - After completion of writing, the faculty discussed the answer and collected the sheets
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO10	PSO1
CO2	3	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO10	PSO1
Justification for correlation	It makes the student to apply the structure concepts they learnt in classroom.	Students will be able to write a program using structures according to the problem	Students will be able to provide solutions to problems which involves handling of heterogeneous data	Students programming skill will be improved as they are replicating the structure concept they have learnt	The recollecting/ remembering and writing skill earned through this activity helps the students in solving problems related to structures

- **Images / Screenshot of the practice:**

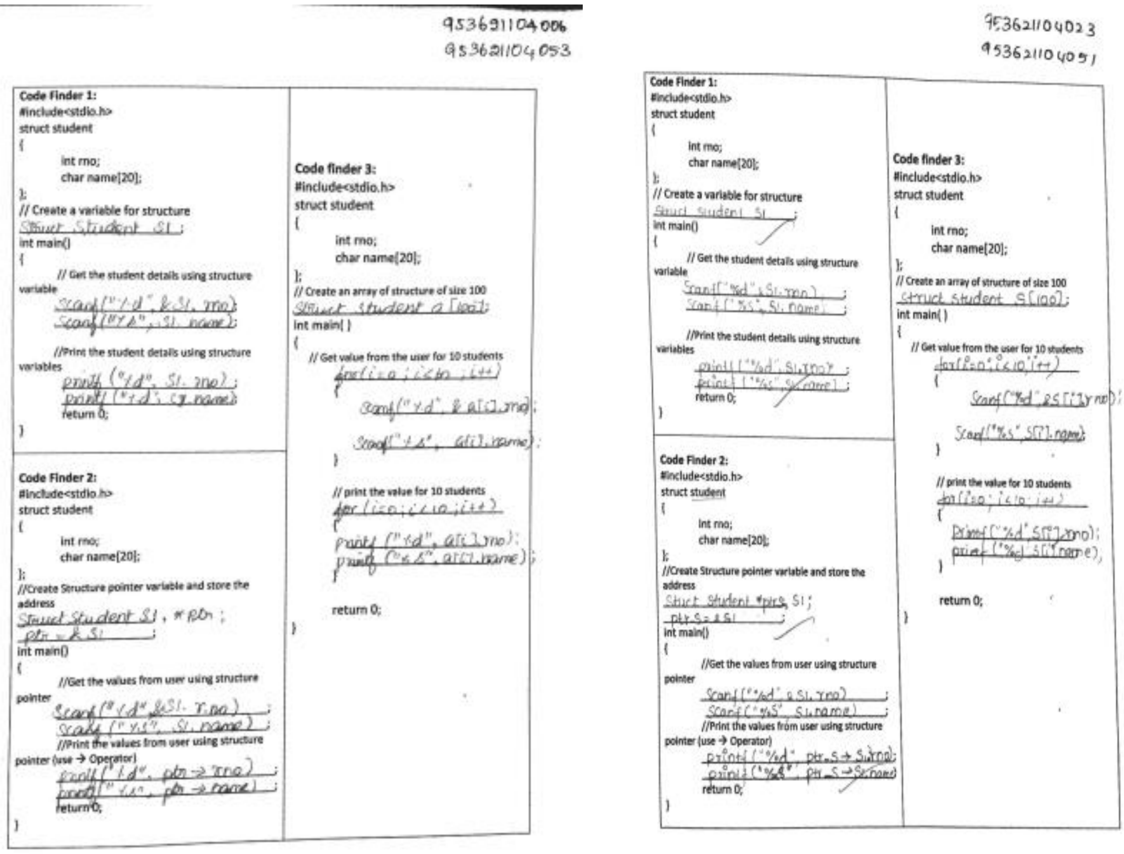

Fig 1: Samples of Code finder activity

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    - Students felt they were able to clearly understand the topic.
    - They said before the happening of activity all the concepts were bit confused. After the activity, the understandability was improved
    - Few students found difficulty in writing program using structure and pointer
  - ***Benefit of the practice:***
    - As the student involves finding the missed code the coding skill will be improved.
    - The entire program will be recollected and remembered
    - The students can share their gained knowledge with their friends
  - ***Challenges faced in implementation:***
    - Due to time constraint, able to give programs on few structure topics. Unable to cover topics on functions and structures, self referential structures etc.
    - Few students took more time to recollect the topic
    - Entire team assessment was difficult within stipulated time.
- **References:**

  - https://www.sololearn.com/Discuss/1663494/fill-in-the-blanks-to-declare-a-struct-and-a-variable-of-that-type
  - https://www.codesdope.com/c-structure/
  - https://corp.kaltura.com/blog/innovative-teaching-strategies/

Signature of Faculty Member **HOD**